

Content

1.	Case Study	3
2.	Problem Statement	4
3.	Exploratory Analysis Using Pandas.....	5
3.1.	Data Loading	5
3.2.	Dimensions and columns.....	6
3.3.	Row preview	15
3.4.	Unique primary key enforcement	19
3.5.	Null Check	21
3.6.	MER.....	23
3.7.	Table cleanup and changes	23
3.8.	Views answering the questions	29
4.	Comparison of Pandas analysis with Spark.....	37
5.	Dashboard on Tableau Public	38
6.	Selected Model.....	39
7.	Model structure in Hive.....	39
8.	Conclusions	41

1. Case Study

This study focuses on analyzing a structured dataset related to professional soccer, compiling information on matches, players, teams, and detailed statistics. The primary objective is to explore various aspects of athletic performance at both the individual and team levels, as well as to analyze trends in market values and the evolution of major competitions.

The dataset includes records of more than 60,000 matches from various seasons, covering more than 400 clubs and approximately 30,000 players.

It is a rich source of information, ideal for conducting exploratory analyses and visualizations that help answer key questions about performance and the dynamics of professional soccer.

This resource, which is automatically updated weekly, was generated from data extracted from the Transfermarkt website using scraping techniques and processing using Python and SQL. It is a rich source of information, ideal for conducting exploratory analyses and creating visualizations that help answer key questions about the performance and dynamics of professional soccer.

2. Problem Statement

The study of the soccer dataset compiled from Transfermarkt aims to answer relevant questions about the performance of players, clubs, and competitions. Using this structured database, it is possible to identify patterns, analyze trends in various indicators over time, and gain deeper insights into the workings of professional soccer.

Below are the 8 questions selected to guide the analysis:

1. Which are the top 10 teams with the most matches, and what is their offensive efficiency?
KPI: offensive efficiency = goals scored ÷ matches played
2. What are the top 10 clubs with the largest net losses? KPI:
total balance = revenue - expenses
3. How many players are there per country by nationality, and how are they distributed on a global heat map?
4. How has the number of goals per season evolved over time in the five major European leagues?
5. What is the average market value of FC Barcelona players, and how has it changed?
6. What are the differences between the Premier League and the Eredivisie in 2024 in terms of match events such as goals, cards, and substitutions?
7. Which 10 teams have made the most transfers over the years?
8. How has the average age of players by position changed over the years in LaLiga?

3. Exploratory analysis using Pandas

During the initial stage of analyzing the soccer dataset obtained from Transfermarkt, the Pandas library was used as the primary tool to explore and understand the available information. This tool proved essential for inspecting the structure of the files, performing filtering, grouping, and transformations, as well as for establishing relationships between different entities, such as players, clubs, matches, and statistics. Thanks to its versatility and ease of use, it was possible to efficiently handle large volumes of data, allowing for agile and organized progress in the exploratory analysis.

3.1. Data Loading

As a starting point for the analysis, data was read from CSV files containing information related to professional soccer.

An automated process was implemented to scan and load all necessary files, using the `pandas.read_csv()` function with UTF-8 encoding. This encoding was essential to ensure the correct reading of special characters in names and attributes, avoiding interpretation errors or incorrect symbols.

The parameter `na_values=["", "NA", "null"]` was used to identify missing values and treat them as NaN, which improves data quality. Additionally, potential errors during loading were handled, and column names were standardized. Column names were standardized by converting them to lowercase, removing spaces, and replacing them with underscores. This normalization simplifies access to the columns during analysis, prevents common errors, and improves consistency in the entire dataset.

Using a dictionary to store all DataFrames enables centralized and flexible access to each table, facilitating data integration and subsequent exploration.

3.2. Dimensions and Columns

A preliminary analysis was conducted on the selected original tables to explore their structure before applying data cleaning processes. At this stage, the number of records (rows), number of attributes (columns), column names, and the data type present in each were identified, taking into account their meaning within the domain. This procedure allowed us to understand the general structure and the type of information contained in each table. For example, a table with dimensions (120, 12) indicates that it has 120 rows and 12 columns. The column names represent the variables that describe each record, providing a clear idea of the content and its purpose.

df_appearances table:

- **Dimensions:** (1,706,806 rows, 13 columns)
- **Description:** Contains records of all player appearances in matches for their respective clubs.
- **Columns:**
 - appearance_id: Unique identifier for the player's appearance in a match.
 - game_id: Identifier for the match played.
 - player_id: Player identifier.
 - player_club_id: The ID of the club the player belonged to during the match.
 - player_current_club_id: Identifier of the player's current club.
 - date: Date on which the match was played.

- player_name: Player's full name.
- competition_id: Identifier of the competition in which the match was played.
- yellow_cards: Number of yellow cards received.
- red_cards: Number of red cards received.
- goals: Number of goals scored by the player.
- assists: Number of assists made.
- minutes_played: Minutes played by the player in that match.

df_clubGames table:

- **Dimensions:** (148,052 rows, 11 columns)
- **Description:** Contains records of each club's individual performance in every match, including goals scored, position, coach, home status, result, and opponent information.
- **Columns:**
 - game_id: Unique identifier for the match.
 - club_id: Identifier of the main club to which the row refers.
 - own_goals: Goals scored by the main club in the match.
 - own_position: The main club's position in the standings at the time of the match.
 - own_manager_name: Name of the main club's manager.
 - opponent_id: Identifier of the opposing club.
 - opponent_goals: Goals scored by the opposing club in the match.
 - opponent_position: The opposing club's position in the standings at the time of the match.
 - opponent_manager_name: Name of the opposing club's manager.
 - hosting: Indicates whether the home team was the host
 - is_win: Indicates whether the home club won the match

df_clubs tables:

- **Dimensions:** (439 rows, 17 columns)
- **Description:** Contains detailed information about soccer clubs, including general information, roster, stadium, coaching staff, history, and market value.
- **Columns:**
 - club_id: Unique identifier for the club.
 - club_code: Club code/abbreviation
 - name: Full name of the club.
 - domestic_competition_id: Identifier of the national league in which the club competes.
 - total_market_value: Market value of the club.
 - squad_size: Total number of players on the roster.
 - average_age: Average age of the players on the roster.
 - foreigners_number: Number of foreign players on the team.
 - foreigners_percentage: Percentage of foreign players out of the total squad.
 - national_team_players: Number of players who also play for national teams.
 - stadium_name: Name of the stadium where the club plays its home games.
 - stadium_seats: Total seating capacity of the stadium.
 - net_transfer_record: The club's net transfer balance.
 - coach_name: Name of the club's head coach.
 - last_season: The most recent season for which data is available for the club.
 - filename: Path to the source file containing the club's original data extracted from Transfermarkt.
 - url: Web link with more information about the club.

df_competitions tables:

- **Dimensions:** (44 rows, 11 columns)
- **Description:** Contains detailed information about soccer competitions, both domestic and international.
- **Columns:**
 - competition_id: Unique identifier for the competition.
 - competition_code: Key representing the competition.
 - name: Full name of the competition.
 - sub_type: Competition subtype
 - type: General competition type (e.g., domestic, international, or other)
 - country_id: The identifier for the country where the competition is held.
 - country_name: Name of the country where the competition takes place.
 - domestic_league_code: Code of the related national league.
 - confederation: The continental confederation to which the competition belongs
 - url: Reference link to the data source. (links do not work)
 - is_major_national_league: Indicates whether the competition is a major national league.

df_gameEvents tables:

- **Dimensions:** (1,035,043 rows, 10 columns)
- **Description:** Contains all events recorded during matches, such as goals, assists, substitutions, or cards. Each row represents a specific event that occurred in a given match.
- **Columns:**
 - game_event_id: Unique identifier for each event in the match.
 - date: The date on which the match in which the event occurred was played.

- game_id: Identifier of the match associated with the event.
- minute: Minute of the match in which the event occurred.
- type: Type of event (e.g., goal, assist, substitution, yellow card, etc.).
- club_id: Identifier of the club to which the player involved in the event belongs.
- player_id: Identifier of the main player involved in the event.
- description: Textual description of the event, possibly with additional details. (Try to resolve this)
- player_in_id: In the case of substitutions, the ID of the player entering the field.
- player_assist_id: In the case of goals, the ID of the player who provided the assist (if applicable).

Tables df_gameLinesUp:

- **Dimensions:** (2,191,911 rows, 10 columns)
- **Description:** Contains all events recorded during matches, such as goals, assists, substitutions, or cards. Each row represents a specific event that occurred in a given match.
- **Columns:**
 - game_lineups_id: Unique identifier for the lineup record.
 - date: Date on which the match was played.
 - game_id: Match identifier.
 - player_id: Identifier of the player in the lineup.
 - club_id: Identifier of the club to which the player belongs in that match.
 - player_name: Player's full name.
 - type: Indicates whether the player started the match or came off the bench

- position: Player's position on the field.
- number: The player's jersey number for that game.
- team_captain: Indicates whether the player was the team captain.

df_games tables:

- **Dimensions:** (74,026 rows, 23 columns)
- **Description:** Contains all events recorded during matches, such as goals, assists, substitutions, or cards. Each row represents a specific event that occurred in a given match.
- **Columns:**
 - game_id: Unique identifier for the match.
 - competition_id: ID of the competition to which the match belongs.
 - season: The season in which the match was played.
 - round: Matchday of the tournament.
 - date: Date the match was played.
 - home_club_id: Home team ID.
 - away_club_id: ID of the visiting club.
 - home_club_goals: Goals scored by the home team.
 - away_club_goals: Goals scored by the visiting club.
 - home_club_position: Home team's position in the standings before the match.
 - away_club_position: The visiting club's position in the standings before the match.
 - home_club_manager_name: Name of the home team's head coach.
 - away_club_manager_name: Name of the visiting team's head coach.
 - stadium: Name of the stadium where the match was played.
 - attendance: Number of spectators in attendance.
 - referee: Name of the match referee.

- url: Detailed match information.
- home_club_formation: Home team's tactical formation.
- away_club_formation: Tactical formation of the visiting team.
- home_club_name: Name of the home team.
- away_club_name: Name of the visiting team.
- aggregate: Aggregate score for two-legged matches.
- competition_type: Competition type.

df_playerValuations tables:

- **Dimensions:** (496,606 rows, 5 columns)
- **Description:** Contains historical records of players' market values, including the club they were with and the national competition at that time.
- **Columns:**
 - player_id: Unique player identifier.
 - date: Date of the market value record.
 - market_value_in_eur: Player's market value in euros.
 - current_club_id: The club the player was with on that date.
 - player_club_domestic_competition_id: Identifier of the domestic competition in which the player's club was participating.

df_players tables:

- **Dimensions:** (32,601 rows, 23 columns)
- **Description:** Contains general, contractual, and sporting information about soccer players, including personal details, their current club, and market value.
- **Columns:**
 - player_id: Unique identifier for the player.
 - first_name: Player's first name.
 - last_name: Player's last name.
 - name: Player's full name or common name.
 - last_season: The player's most recent season recorded in the database.
 - current_club_id: Identifier of the player's current club.
 - player_code: Additional player identifier (possibly from the source site).
 - country_of_birth: Player's country of birth.
 - city_of_birth: Player's city of birth.
 - country_of_citizenship: The player's nationality or citizenship.
 - date_of_birth: Date of birth.
 - sub_position: Specific sub-position on the field.
 - position: Specific position on the field.
 - foot: Dominant foot.
 - height_in_cm: Player's height in centimeters.
 - contract_expiration_date: Expiration date of the contract with the current club.
 - agent_name: Name of the player's agent.
 - image_url: Link to the player's image.
 - url: Link to the player's profile.
 - current_club_domestic_competition_id: ID of the current club's domestic league.
 - current_club_name: Name of the current club.

- `market_value_in_eur`: Player's current market value in euros.
- `highest_market_value_in_eur`: Highest market value reached by the player in euros.

df_transfers tables:

- **Dimensions:** (79,646 rows, 10 columns)
- **Description:** Contains detailed information on player transfers between clubs, such as player withdrawals, including dates, amounts paid, market value, and the names of the clubs involved.
- **Columns:**
 - `player_id`: Unique identifier of the transferred player.
 - `transfer_date`: Exact date the transfer took place.
 - `transfer_season`: Season in which the transfer occurred.
 - `from_club_id`: Identifier of the originating club.
 - `to_club_id`: ID of the destination club.
 - `from_club_name`: Name of the originating club.
 - `to_club_name`: Name of the destination club.
 - `transfer_fee`: Amount paid for the transfer.
 - `market_value_in_eur`: Market value of the player at the time of the transfer.
 - `player_name`: Name of the transferred player.

3.3. Preview rows

df_appearances table

	player_id	first_name	last_name	name	last_season	current_club_id	player_code	country_of_birth	city_of_birth	country_of_citizenship	date_of_birth
0	10	Miroslav	Klose	Miroslav Klose	2015	398	miroslav-klose	Poland	Opole	Germany	1978-06-09 00:00:00
1	26	Roman	Weidenfeller	Roman Weidenfeller	2017	16	roman-weidenfeller	Germany	Diez	Germany	1980-08-06 00:00:00
2	65	Dimitar	Berbatov	Dimitar Berbatov	2015	1091	dimitar-berbatov	Bulgaria	Blagoevgrad	Bulgaria	1981-01-30 00:00:00
3	77	NaN	Lúcio	Lúcio	2012	506	lucio	Brazil	Brasília	Brazil	1978-05-08 00:00:00
4	80	Tom	Starke	Tom Starke	2017	27	tom-starke	East Germany (GDR)	Freital	Germany	1981-03-18 00:00:00

sub_position	position	foot	height_in_cm	contract_expiration_date	agent_name	image_url
Centre-Forward	Attack	right	184.0		NaN	ASBW Sport Marketing https://img.a.transfermarkt.technology/portrai...
Goalkeeper	Goalkeeper	left	190.0		NaN	Neubauer 13 GmbH https://img.a.transfermarkt.technology/portrai...
Centre-Forward	Attack	NaN	NaN		NaN	CSKA-AS-23 Ltd. https://img.a.transfermarkt.technology/portrai...
Centre-Back	Defender	NaN	NaN		NaN	NaN https://img.a.transfermarkt.technology/portrai...
Goalkeeper	Goalkeeper	right	194.0		NaN	IFM https://img.a.transfermarkt.technology/portrai...

url	current_club_domestic_competition_id	current_club_name	market_value_in_eur	highest_market_value_in_eur
https://www.transfermarkt.co.uk/miroslav-klose...	IT1	Società Sportiva Lazio S.p.A.	1000000.0	30000000.0
https://www.transfermarkt.co.uk/roman-weidenfe...	L1	Borussia Dortmund	750000.0	8000000.0
https://www.transfermarkt.co.uk/dimitar-berbat...	GR1	Panthessalonikios Athlitikos Omilos Konstantin...	1000000.0	34500000.0
https://www.transfermarkt.co.uk/lucio/profil/s...	IT1	Juventus Football Club	200000.0	24500000.0
https://www.transfermarkt.co.uk/tom-starke/pro...	L1	FC Bayern München	100000.0	3000000.0

df_clubGames table

	game_id	club_id	own_goals	own_position	own_manager_name	opponent_id	opponent_goals	opponent_position	opponent_manager_name	hosting	is_win
0	2320450	1468.0	0.0	NaN	Holger Bachthaler	24.0	2.0	NaN	Armin Veh	Home	0
1	2320454	222.0	0.0	NaN	Volkan Uluc	79.0	2.0	NaN	Bruno Labbadia	Home	0
2	2320460	1.0	3.0	NaN	Jürgen Luginger	86.0	1.0	NaN	Robin Dutt	Home	1
3	2320472	2036.0	4.0	NaN	Frank Schmidt	72.0	5.0	NaN	Alexander Schmidt	Home	0
4	2321027	33.0	3.0	8.0	Jens Keller	41.0	3.0	9.0	Thorsten Fink	Home	0

df_clubs table

	club_id	club_code	name	domestic_competition_id	total_market_value	squad_size	average_age	foreigners_number	foreigners_percentage
0	105	sv-darmstadt-98	SV Darmstadt 98	L1	NaN	27	25.6	13	48.1
1	11127	ural-ekaterinburg	Ural Yekaterinburg	RU1	NaN	30	26.5	11	36.7
2	114	besiktas-istanbul	Beşiktaş Jimnastik Kulübü	TR1	NaN	30	26.6	15	50.0
3	12	as-rom	Associazione Sportiva Roma	IT1	NaN	26	26.3	18	69.2
4	148	tottenham-hotspur	Tottenham Hotspur Football Club	GB1	NaN	30	25.5	21	70.0

national_team_players	stadium_name	stadium_seats	net_transfer_record	coach_name	last_season	filename	url
1	Merck-Stadion am Böllenfalltor	17810	+€3.05m	NaN	2023	./data/raw/transfermarkt-scraper/2023/clubsj...	https://www.transfermarkt.co.uk/sv-darmstadt-9...
3	Yekaterinburg Arena	23000	+€880k	NaN	2023	./data/raw/transfermarkt-scraper/2023/clubsj...	https://www.transfermarkt.co.uk/ural-ekaterinb...
8	Beşiktaş Park	42445	€-25.26m	NaN	2024	./data/raw/transfermarkt-scraper/2024/clubsj...	https://www.transfermarkt.co.uk/besiktas-istan...
17	Olimpico di Roma	70634	€-76.90m	NaN	2024	./data/raw/transfermarkt-scraper/2024/clubsj...	https://www.transfermarkt.co.uk/as-rom/startse...
18	Tottenham Hotspur Stadium	62850	€-120.05m	NaN	2024	./data/raw/transfermarkt-scraper/2024/clubsj...	https://www.transfermarkt.co.uk/tottenham-hot...

df_competitions table

	competition_id	competition_code	name	sub_type	type	country_id	country_name	domestic_league_code	confederation
0	CIT	italy-cup	italy-cup	domestic_cup	domestic_cup	75	Italy	IT1	europa
1	NLSC	johan-crujff-schaal	johan-crujff-schaal	domestic_super_cup	other	122	Netherlands	NL1	europa
2	GRP	kypello-elladas	kypello-elladas	domestic_cup	domestic_cup	56	Greece	GR1	europa
3	POSU	supertaca-candido-de-oliveira	supertaca-candido-de-oliveira	domestic_super_cup	other	136	Portugal	PO1	europa
4	RUSS	russian-super-cup	russian-super-cup	domestic_super_cup	other	141	Russia	RU1	europa

url	is_major_national_league
https://www.transfermarkt.co.uk/italy-cup/star...	False
https://www.transfermarkt.co.uk/johan-crujff-...	False
https://www.transfermarkt.co.uk/kypello-ellada...	False
https://www.transfermarkt.co.uk/supertaca-cand...	False
https://www.transfermarkt.co.uk/russian-super-...	False

df_gameEvents table

	game_event_id	date	game_id	minute	type	club_id	player_id		description	player_in_id	player_assist_id
0	2f41da30c471492e7d4a984951671677	2012-08-05	2211607	77	Cards	610	4425	1.	Yellow card , Mass confrontation	NaN	NaN
1	a72f7186d132775f234d3e2f7bc0ed5b	2012-08-05	2211607	77	Cards	383	33210	1.	Yellow card , Mass confrontation	NaN	NaN
2	b2d721eae4692a5c59a92323689ef18	2012-08-05	2211607	3	Goals	383	36500	, Header, 1. Tournament Goal Assist: , Corner,...		NaN	56416.0
3	aef768899cedac0c9a650980219075a2	2012-08-05	2211607	53	Goals	383	36500	, Right-footed shot. 2. Tournament Goal Assist...		NaN	146258.0
4	5d6d9533023057b6619ecd145a038bbe	2012-08-05	2211607	74	Substitutions	383	36500		, Not reported	49499.0	NaN

df_gameLinesUp table

	game_lineups_id	date	game_id	player_id	club_id	player_name	type	position	number	team_captain
0	b2dbe01c3656b06c8e23e9de714e26bb	2013-07-27	2317258	1443	610	Christian Poulsen	substitutes	Defensive Midfield	5	0
1	b50a3ec6d52fd1490aab42042ac4f738	2013-07-27	2317258	5017	610	Niklas Moisander	starting_lineup	Centre-Back	4	0
2	7d890e6d0ff8af84b065839966a0ec81	2013-07-27	2317258	9602	1090	Maarten Martens	substitutes	Left Winger	11	0
3	8c355268678b9bbc7084221b1f0fde36	2013-07-27	2317258	12282	610	Daley Blind	starting_lineup	Left-Back	17	0
4	76193074d549e5fdce4cdcbba0d66247	2013-07-27	2317258	25427	1090	Roy Beerens	starting_lineup	Right Winger	23	0

Table df_games

	game_id	competition_id	season	round	date	home_club_id	away_club_id	home_club_goals	away_club_goals	home_club_position	away_club_position
0	2321027	L1	2013	1. Matchday	2013-08-11	33.0	41.0	3.0	3.0	8.0	9.0
1	2321033	L1	2013	1. Matchday	2013-08-10	23.0	86.0	0.0	1.0	13.0	7.0
2	2321044	L1	2013	2. Matchday	2013-08-18	16.0	23.0	2.0	1.0	1.0	15.0
3	2321060	L1	2013	3. Matchday	2013-08-25	23.0	24.0	0.0	2.0	18.0	11.0
4	2321072	L1	2013	5. Matchday	2013-09-14	16.0	41.0	6.0	2.0	1.0	15.0

home_club_manager_name	away_club_manager_name	stadium	attendance	referee	url	home_club_formation
Jens Keller	Thorsten Fink	Veltins-Arena	61973.0	Manuel Gräfe	https://www.transfermarkt.co.uk/fc-schalke-04_...	4-2-3-1
Torsten Lieberknecht	Robin Dutt	EINTRACHT-Stadion	23000.0	Deniz Aytekin	https://www.transfermarkt.co.uk/eintracht-brau...	4-3-2-1
Jürgen Klopp	Torsten Lieberknecht	SIGNAL IDUNA PARK	80200.0	Peter Sippel	https://www.transfermarkt.co.uk/borussia-dortm...	4-2-3-1
Torsten Lieberknecht	Armin Veh	EINTRACHT-Stadion	23325.0	Wolfgang Stark	https://www.transfermarkt.co.uk/eintracht-brau...	4-3-2-1
Jürgen Klopp	Thorsten Fink	SIGNAL IDUNA PARK	80645.0	Tobias Welz	https://www.transfermarkt.co.uk/borussia-dortm...	4-2-3-1

away_club_formation	home_club_name	away_club_name	aggregate	competition_type
4-2-3-1	FC Schalke 04	Hamburger SV	3:3	domestic_league
4-3-1-2	Eintracht Braunschweig	Sportverein Werder Bremen von 1899	0:1	domestic_league
4-3-2-1	Borussia Dortmund	Eintracht Braunschweig	2:1	domestic_league
4-2-3-1	Eintracht Braunschweig	Eintracht Frankfurt Fußball AG	0:2	domestic_league
3-5-2	Borussia Dortmund	Hamburger SV	6:2	domestic_league

df_playerValuations table

	player_id	date	market_value_in_eur	current_club_id	player_club_domestic_competition_id
0	405973	2000-01-20	150000	3057	BE1
1	342216	2001-07-20	100000	1241	SC1
2	3132	2003-12-09	400000	126	TR1
3	6893	2003-12-15	900000	984	GB1
4	10	2004-10-04	7000000	398	IT1

df_players table

	player_id	first_name	last_name	name	last_season	current_club_id	player_code	country_of_birth	city_of_birth	country_of_citizenship	date_of_birth
0	10	Miroslav	Klose	Miroslav Klose	2015	398	miroslav-klose	Poland	Opole	Germany	1978-06-09 00:00:00
1	26	Roman	Weidenfeller	Roman Weidenfeller	2017	16	roman-weidenfeller	Germany	Diez	Germany	1980-08-06 00:00:00
2	65	Dimitar	Berbatov	Dimitar Berbatov	2015	1091	dimitar-berbatov	Bulgaria	Blagoevgrad	Bulgaria	1981-01-30 00:00:00
3	77	NaN	Lúcio	Lúcio	2012	506	lucio	Brazil	Brasília	Brazil	1978-05-08 00:00:00
4	80	Tom	Starke	Tom Starke	2017	27	tom-starke	East Germany (GDR)	Freital	Germany	1981-03-18 00:00:00

sub_position	position	foot	height_in_cm	contract_expiration_date	agent_name	image_url
Centre-Forward	Attack	right	184.0	NaN	ASBW Sport Marketing	https://img.a.transfermarkt.technology/portrai...
Goalkeeper	Goalkeeper	left	190.0	NaN	Neubauer 13 GmbH	https://img.a.transfermarkt.technology/portrai...
Centre-Forward	Attack	NaN	NaN	NaN	CSKA-AS-23 Ltd.	https://img.a.transfermarkt.technology/portrai...
Centre-Back	Defender	NaN	NaN	NaN	NaN	https://img.a.transfermarkt.technology/portrai...
Goalkeeper	Goalkeeper	right	194.0	NaN	IFM	https://img.a.transfermarkt.technology/portrai...

	uri	current_club_domestic_competition_id	current_club_name	market_value_in_eur	highest_market_value_in_eur
	https://www.transfermarkt.co.uk/miroslav-klose...	IT1	Società Sportiva Lazio S.p.A.	1000000.0	30000000.0
	https://www.transfermarkt.co.uk/roman-weidenfe...	L1	Borussia Dortmund	750000.0	8000000.0
	https://www.transfermarkt.co.uk/dimitar-berbat...	GR1	Panthessalonikios Athlitikos Omilos Konstantin...	1000000.0	34500000.0
	https://www.transfermarkt.co.uk/lucio/profil/s...	IT1	Juventus Football Club	200000.0	24500000.0
	https://www.transfermarkt.co.uk/tom-starke/pro...	L1	FC Bayern München	100000.0	3000000.0

Table df_transfers

	player_id	transfer_date	transfer_season	from_club_id	to_club_id	from_club_name	to_club_name	transfer_fee	market_value_in_eur	player_name
0	16136	2026-07-01	26/27	417	123	OGC Nice	Retired	NaN	500000.0	Dante
1	1138758	2026-07-01	26/27	336	631	Sporting CP	Chelsea	52140000.0	45000000.0	Geovany Quenda
2	195778	2026-06-30	25/26	79	27	VfB Stuttgart	Bayern Munich	0.0	12000000.0	Alexander Nübel
3	569033	2026-06-30	25/26	39	27	1.FSV Mainz 05	Bayern Munich	0.0	4000000.0	Armino Sieb
4	626913	2026-06-30	25/26	398	380	Lazio	Salernitana	0.0	15000000.0	Boulaye Dia

3.4. Checking for unique primary keys

From this point forward, in the analysis we will be working with the tables after having cleaned them. This allows us to ensure correct processing both to complete the proposed analysis and the Viewing graphs by answering the questions. As can be seen, neither the tables with simple keys nor those with composite keys contain duplicate unique keys, which ensures data integrity and prevents future inconsistencies.

```

=====player_id=====
Empty DataFrame
Columns: [player_id, count]
Index: []
=====

=====club_id=====
Empty DataFrame
Columns: [club_id, count]
Index: []
=====

=====game_id=====
Empty DataFrame
Columns: [game_id, count]
Index: []
=====

=====competition_id=====
Empty DataFrame
Columns: [competition_id, count]
Index: []
=====

=====game_lineups_id=====
Empty DataFrame
Columns: [game_lineups_id, count]
Index: []
=====

=====df_appearances=====
Empty DataFrame
Columns: [game_id, player_id, count]
Index: []
=====

=====df_clubGames=====
Empty DataFrame
Columns: [game_id, club_id, opponent_id, count]
Index: []
=====

=====df_playerValuations=====
Empty DataFrame
Columns: [player_id, date, count]
Index: []
=====

=====df_transfers=====
Empty DataFrame
Columns: [player_id, transfer_date, from_club_id, to_club_id, count]
Index: []
=====

```

3.5. Checking for Null Values

At this stage of the analysis, the presence of null values in the different tables was reviewed to understand their impact and determine the most appropriate treatment. The reasoning followed is detailed below:

- **df_clubs table**

The variable `net_transfer_balance_million` contains 100 null values. Upon analyzing their context, it was determined that these correspond to clubs of lesser importance or with low financial participation; therefore, it was decided not to filter or impute them. Given that their volume is so small, they do not affect the overall transfer analyses or the visualization of the information.

- **df_gameEvents table**

In this case, there are many null values in the columns `player_in_id` (537,365 nulls) and `player_assist_id` (878,284 nulls). This behavior is expected and natural given the definition of these variables:

- `player_in_id` is used only in substitution events, so most events (goals, cards, fouls, etc.) do not contain this information, resulting in justified null values.
- `player_assist_id` refers to the player who provides the assist on a goal. However, not all goals result from an assist (for example, a penalty kick or a goal following a rebound), which explains the high proportion of null values in this column.

In both cases, these nulls are inherent to the very nature of match events, so they are not considered problematic.

- **df_players table**

The `date\_of\_birth` variable contains 47 missing values, which represents an extremely small subset relative to the total size of the table. Since the impact is minimal and does not compromise the overall integrity of the analysis, it was decided to retain these records without imputing or deleting them.

- **df_transfers table**

The market_value_in_eur column contains 30,316 null values. This variable reflects the player's market value at the time of the transfer. However, not all transfers necessarily involve necessarily a monetary payment or an estimated value (e.g., loan deals, retirements from soccer, or transfers at no cost). This explains the high proportion of nulls, since in such cases no associated market value is recorded. Therefore, it is considered appropriate to retain these null values, as they stem from the very nature of the data.

Analysis of null values shows that most missing fields are due to the very nature of the information, such as the absence of assists on certain goals or the absence of an incoming player in events that do not correspond to substitutions. In other cases, the null values represent a minimal percentage of the total dataset, without compromising the consistency or reliability of the data. For this reason, it was decided not to apply any additional transformations or filters, preserving the integrity and consistency of the dataset to ensure that the analysis and visualizations accurately reflect reality. No null values were detected in the other tables, so no further action was required for them.

3.6. MER



3.7. Table cleaning and changes

- **df_appearances table:**

The df_appearances dataframe was cleaned by converting the date column to datetime format for proper time handling, and by removing unnecessary columns such as player_name and player_current_club_id to optimize storage. Duplicate records were verified and removed based on the combination of game_id and player_id, thus ensuring the uniqueness of appearances per match and player. Additionally, the relevant numeric columns yellow_cards, red_cards, goals, assists, and minutes_played

by converting them to the Int64 type, which supports null values. To ensure data quality of the data, records with minutes played exceeding 120 were filtered out, as these are considered outliers. Missing values for cards, goals, and assists were also imputed with zero to accurately reflect the absence of such events.

- **Table df_clubGames**

The df_clubGames dataframe was cleaned by converting key identifiers such as game_id, club_id, and opponent_id to the Int64 type to handle nulls more efficiently. Irrelevant columns such as team standings and coach names for both teams were removed, thereby simplifying the table to focus on match results. Additionally, the goal fields (own_goals, opponent_goals) were converted to the Int64 data type, and the hosting and is_win columns were converted to the category and bool, respectively, optimizing storage and semantic clarity. Records with null values in key columns such as club_id, opponent_id, own_goals, and opponent_goals were removed to ensure data integrity.

- **df_clubs table**

Several key transformations were performed on the df_clubs dataframe: the club_id column was cast to Int64 to support null values, and the domestic_competition_id variable was adjusted to the category type to optimize storage. Several numeric columns (squad_size, average_age, etc.) were converted to numeric values, handling conversion errors and imputing missing values in average_age with its median to improve consistency. Additionally, the last_season column was normalized to the Int64 type. For the net_transfer_record column, a transformed field named net_transfer_balance_million was created, expressed in millions of euros to facilitate comparative analysis, subsequently removing the

along with other irrelevant columns such as url, coach_name, filename, total_market_value, foreigners_percentage, and club_code to clean up the table.

- **Table df_competitions**

In the df_competitions dataframe, the value 'international' was assigned to null records in the domestic_league_code column, since they corresponded to international competitions such as the Champions League, thus avoiding missing values and facilitating subsequent segmentation. Irrelevant columns such as url, country_name, type, and competition_code were also removed to simplify the dataset, and sub_type was renamed to type to unify the nomenclature. Finally, the key columns confederation and type were transformed to the category data type, optimizing space and improving the handling of categorical data.

- **df_gameEvents table**

In the dataframe, the `game\_event\_id` column was converted to a string type to ensure consistent identifiers. The `date` column was converted to a datetime format to facilitate temporal analysis and date operations. Key identifiers such as club_id, player_id, player_in_id, and player_assist_id were cast to Int64 for better handling of null values. Additionally, the type column was converted to categorical to optimize space and event analysis. The description was normalized using a custom function that groups and standardizes frequent values such as cards, goals, substitutions, fouls, injuries, among others, leaving the rest as "Other" for greater clarity and consistency. Finally, description was also typed as category for more efficient data handling.

- **df_gameLinesUp table**

In the dataframe, the `game_lineups_id` column was transformed to a string type to maintain consistency in the identifiers. The `date` column was converted to datetime format to facilitate chronological analysis. The `game_id`, `player_id`, and `club_id` were defined as `Int64` to allow for null values and maintain data integrity.

The "type" and "position" columns were converted to categorical variables to optimize memory usage and speed up segmentation operations. For the "position" column, missing values were imputed with the "Unknown" category to avoid incomplete records.

A position mapping was applied to group soccer roles into broader categories such as *Forward*, *Midfielder*, *Defender*, and *Goalkeeper*, facilitating subsequent analysis. Values that did not match the original mapping were left unchanged. Subsequently, the original position column was removed, and the normalized column was renamed to preserve standard nomenclature. Unnecessary columns such as `number` and `team_captain` were also removed to simplify the dataset.

- **df_games table**

In the dataframe, columns considered unnecessary—such as `home_club_position`, `away_club_position`, `home_club_manager_name`, `away_club_manager_name`, `referee`, `url`, `home_club_name`, `away_club_name`, `home_club_formation`, `away_club_formation`, `attendance`, and `stadium` were removed to simplify the dataset structure and retain only the information relevant to sports analysis.

The ``date`` column was converted to the ``datetime`` data type to ensure proper handling of dates, and the primary key identifiers ``game_id``, ``home_club_id``, and ``away_club_id`` were defined as ``Int64``, which ensures support for null values while maintaining the numerical integrity of the data.

Furthermore, columns such as `competition_id`, `season`, `round`, and `competition_type` were converted to the category data type to optimize space and improve performance in groupings and filtering.

Additionally, records with null values in critical club identification fields (`home` and `away`) were removed to ensure the completeness of match information. Finally, matches with dates later than the day of the data load to avoid temporal inconsistencies in the analysis.

- **Table `df_playerValuations`**

In the dataframe, the `date` column was converted to the `datetime` type to ensure proper handling of market valuations over time. The fields `player_id` and `current_club_id` were converted to the `Int64` type to maintain consistency with the possible presence of null values, and the `market_value_in_eur` column was converted to an integer (`Int64`) format to facilitate subsequent economic analysis without losing compatibility with missing values. Additionally, the `player_club_domestic_competition_id` column was removed as it was deemed unnecessary for the analytical purpose, reducing the dimensionality of the dataset. These transformations result in a cleaner, more consistent, and optimized dataframe for future queries or player market analysis models.

- **`df_players` table**

Columns that did not add value to the analysis or were redundant—such as `agent_name`, `country_of_birth`, `city_of_birth`, `image_url`, `url`, `current_club_domestic_competition_id`, `current_club_name`, `player_code`, `first_name`, `last_name`, `contract_expiration_date`, `market_value_in_eur`, and `sub_position`. The `foot` column was standardized by filling in missing values with “Unknown” and converting it to a categorical type.

Identifiers such as `player_id` and `current_club_id`, the `height_in_cm` measure, and the `last_season` variable were converted to the `Int64` type to maintain compatibility with null values. The `date_of_birth` column was converted to datetime format, and the approximate current age in years was calculated by creating a new column named `age`, which was also typed as `Int64`.

Additionally, a boolean column `retired` was defined based on whether the last season played was prior to 2022, allowing retired players who have retired, and individuals with a null value for `height_in_cm` were assigned a value based on the median. Finally, monetary values in the `highest_market_value_in_eur` column were standardized by assigning zero to null values and converting the column to type `Int64`, while the categorical variables `country_of_citizenship` and `position` were normalized by filling in missing values with “Unknown”.

- **Table `df_transfers`**

The data types in the dataframe were standardized to facilitate analysis and handling. The `transfer_date` column was converted to the datetime type, while the identifiers `player_id`, `from_club_id`, and `to_club_id`, as well as the numeric columns

monetary columns `transfer_fee` and `market_value_in_eur` were transformed to the `Int64` type to allow for the correct handling of null values.

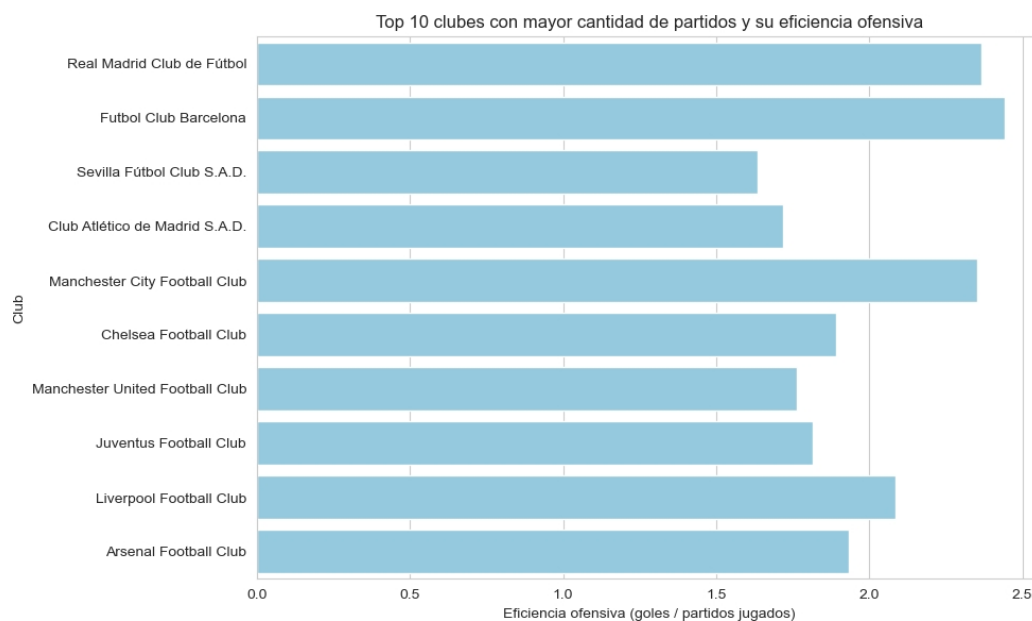
A boolean column named `retired` was created to indicate whether the transfer involves a retired player, identified by the presence of the text “retired” in the destination club’s name. The `transfer_fee` column filled in null values with zero to avoid issues in subsequent calculations, and the `transfer_season` variable was converted to the category type.

Additionally, redundant or sensitive descriptive columns, such as `from_club_name`, `to_club_name`, and `player_name`, were removed to focus on numerical and categorical data.

Finally, a unique incremental identifier, `transfer_id`, was generated for each transfer and placed as the first column to facilitate its use as a primary key.

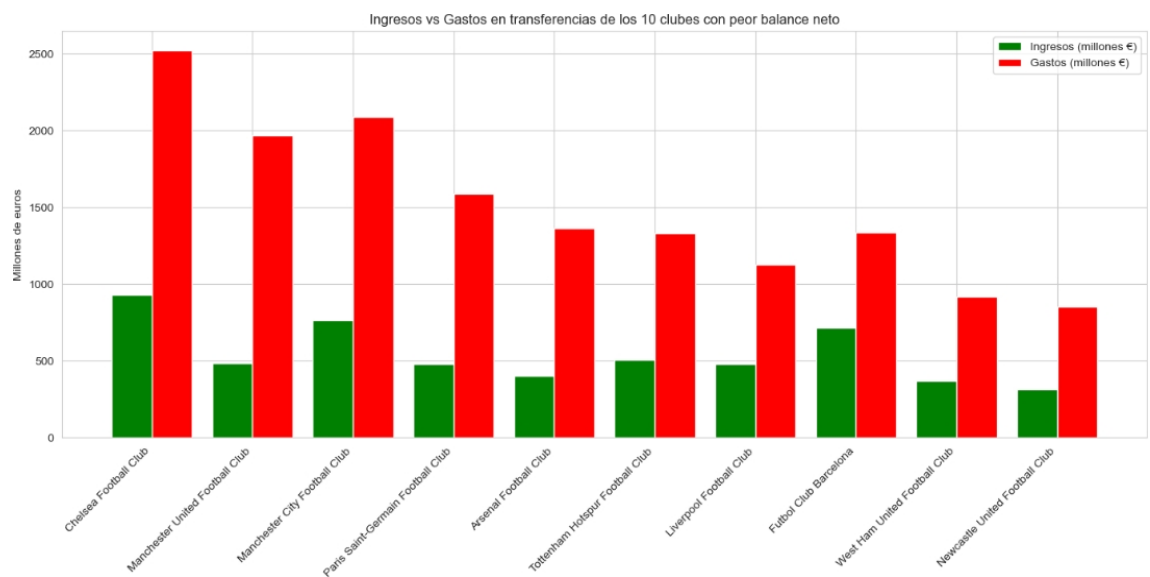
3.8. Visualizations answering the questions

- **What are the top 10 teams with the most matches, and what is their offensive efficiency?**



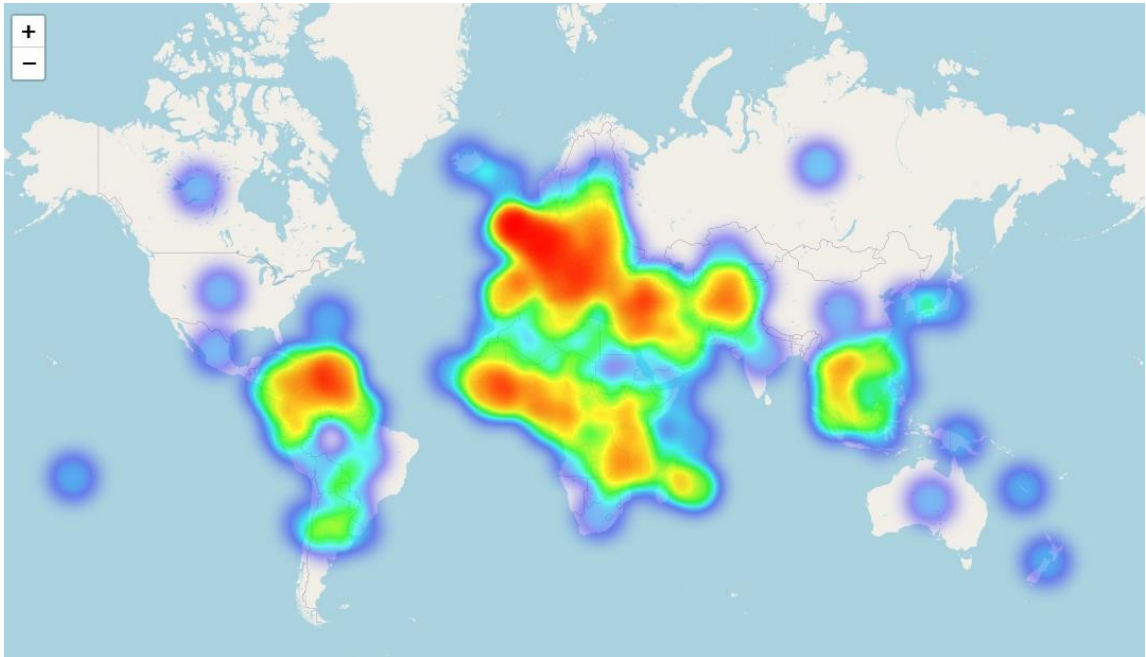
Among the 10 teams with the most matches played, Fútbol Club Barcelona and Real Madrid stand out as the most efficient offensively, averaging approximately 2.3 goals per game. This suggests that, in addition to being teams with a large number of matches played, they also maintain a high offensive output.

- Which are the top 10 clubs with the largest negative goal difference?



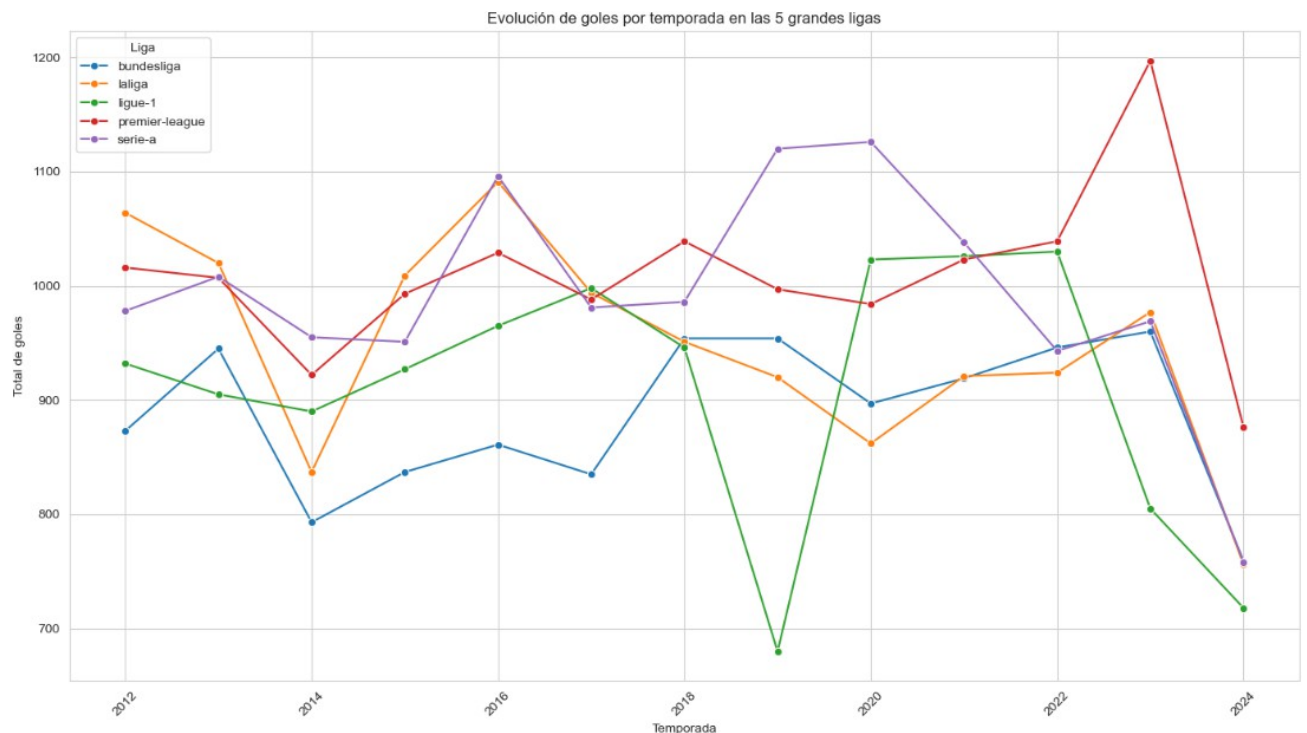
The analysis reveals that Chelsea, Manchester United, and Manchester City lead the ranking of clubs with the largest negative balance in the transfer market. These three teams have adopted an intensive investment strategy, prioritizing the acquisition of talent on a large scale over short-term financial balance.

- **How many players are there per country based on their nationality, and how are they distributed on a global heat map?**



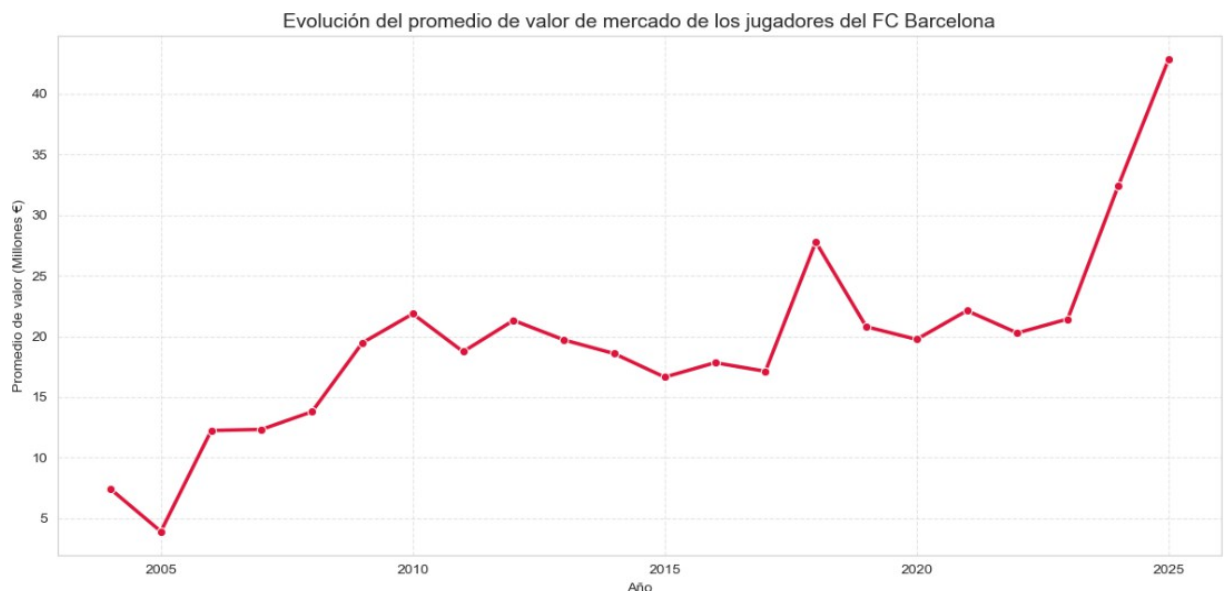
The heat map shows that most professional soccer players come from Europe, South America, and Africa, which are the regions with the highest concentration of soccer talent in the world. In contrast, other regions contribute fewer players or have a much more scattered distribution. This distribution reflects not only each region's soccer tradition but also the development of sports infrastructure, training academies, and opportunities for exporting talent.

- **How has the number of goals per season evolved over time in the five major European leagues?**



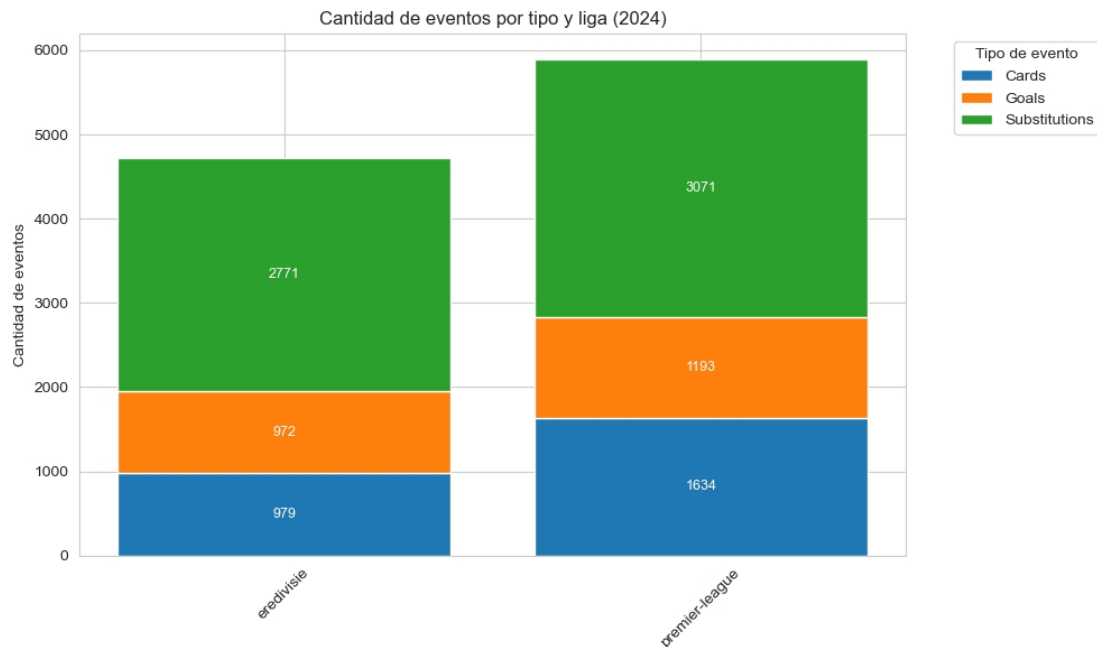
The graph showing how goals per season evolved in the major European leagues (Bundesliga, La Liga, Ligue 1, Premier League, and Serie A) between 2012 and 2024 shows many ups and downs and differences between competitions, without following a clear and consistent trend. Overall, each league had quite year-over-year variation in goals, without a sustained increase or decrease. Serie A and the Premier League typically stand out for having more goals, and Serie A in particular shows very marked ups and downs. Although similar patterns are sometimes observed (such as small drops or recoveries), there are also notable differences; for example, the sharp drop in goals in Ligue 1 around 2019 was not replicated with the same intensity in the other leagues. In recent years, particularly in 2024, most leagues appear to have seen a slight decline in the number of goals following some peaks in previous seasons, thus returning to more typical figures.

- **What is the average market value of FC Barcelona players, and how has it evolved?**



The average market value of FC Barcelona players has shown remarkable resilience and growth. Despite the ups and downs over the decades, the long-term trend is clearly on an upward trend, peaking at a record average value in 2025, suggesting a period of significant appreciation in the value of its squad.

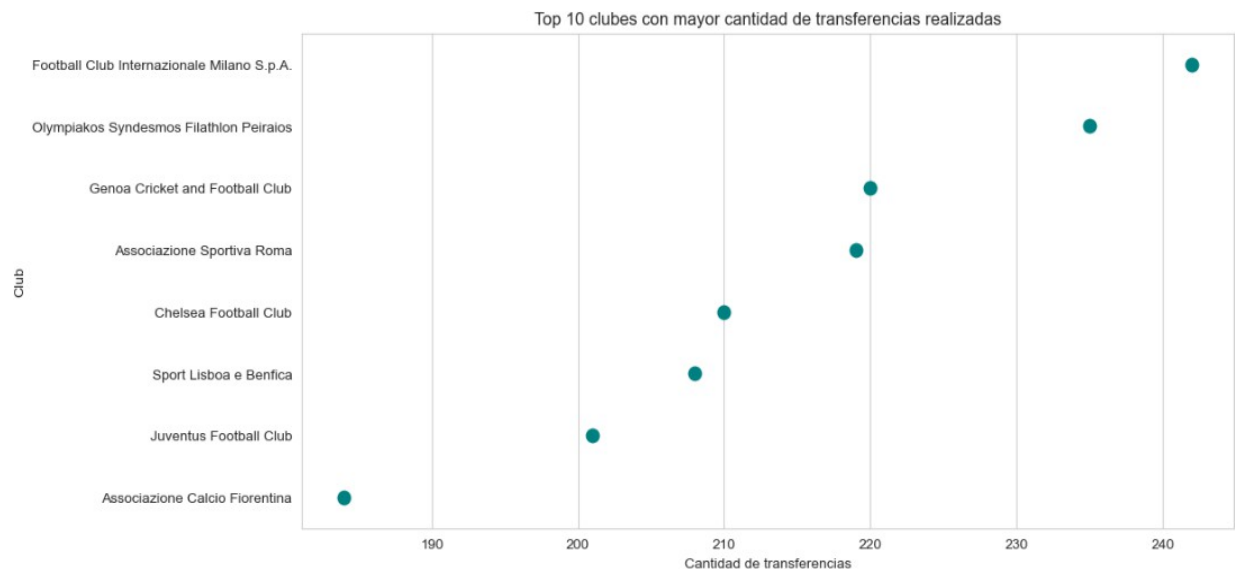
- **What differences exist between the Premier League and the Eredivisie in 2024 in terms of match events such as goals, cards, and substitutions?**



The graph compares the number of match events (cards, goals, and substitutions) between the Eredivisie and the Premier League in 2024, highlighting notable differences in game dynamics. First, the Premier League records a significantly higher total number of events than the Eredivisie, standing out particularly in the number of cards (1,634 versus 979), which could indicate a more physical style of play or stricter refereeing.

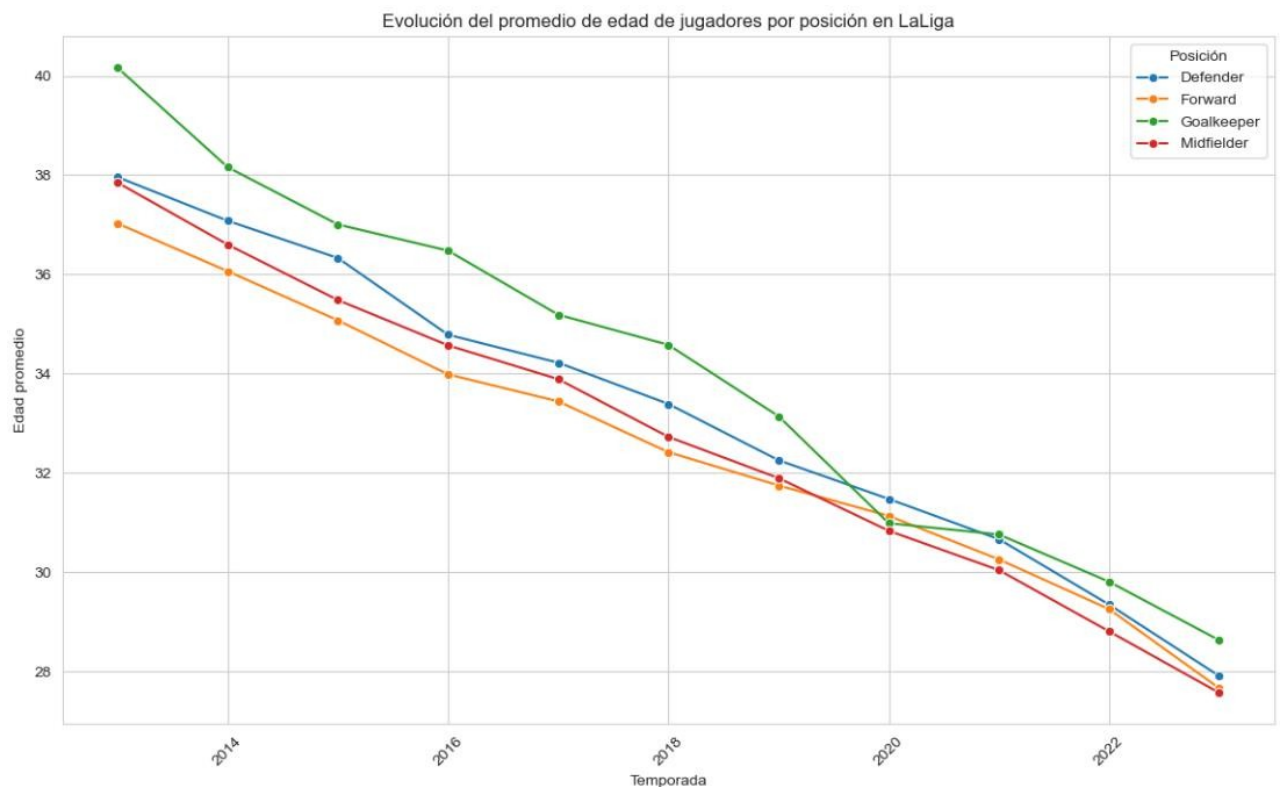
It also outscores the Dutch league in goals (1,193 to 972), reflecting a higher offensive output in the English league. In terms of substitutions, both competitions show high numbers, although the Premier League recorded a slightly higher figure (3,071 versus 2,771), suggesting intense and active squad management. Overall, these data show that in 2024, the Premier League featured a more intense style of play, both physically and offensively, compared to the Eredivisie.

- Which are the 10 teams with the highest number of transfers over the years?



The chart highlights the eight clubs with the highest number of transfers made over the years, highlighting their intense activity in the transfer market. Inter Milan tops the ranking with approximately 242 transfers, closely followed by Olympiacos Piraeus with around 235. Other teams with a high volume of transactions also appear, such as Genoa (220), AS Roma (218), Chelsea (210), Benfica (208), Juventus (201), and Fiorentina (185). The strong presence of Italian clubs among the top spots (Inter, Genoa, Roma, Juventus, and Fiorentina) is striking, which could reflect a trend in Serie A toward frequent rotation and constant renewal of their squads.

- **How has the average age of players by position changed over the years in LaLiga?**



The chart analyzes the trend in the average age of LaLiga players, broken down by position (defenders, forwards, goalkeepers, and midfielders) from 2013 to 2023, and shows a clear trend toward a younger roster in across all positions. Over the period, there was a steady decline in the average age, which was particularly noticeable among goalkeepers, who started out at around 40 years old in 2013 and saw a sharp drop in the early years. Midfielders, who were initially the youngest (around 36 years old), also followed this downward trend.

Furthermore, the graph shows how the age differences between positions, which were quite marked at the beginning, gradually narrowed. Toward the end of the period (2020–2023), all positions converged within a much narrower age range, falling between 27.5 and 30 years old. At the end of the period

Upon analysis, midfielders and forwards stand out as the youngest (around 27.5–28 years old), while defenders and goalkeepers are slightly older, though all remain under 30.

In summary, LaLiga has experienced sustained rejuvenation over the past decade, with increasingly younger rosters and a narrowing gap in the average age across different positions, which could reflect new strategies in talent recruitment and development.

4. Comparison of Pandas Analysis with Spark

In my view, Pandas proved very intuitive for performing data transformations and cleaning, thanks to its straightforward approach and the availability of predefined functions that allow you to move forward quickly without too many intermediate steps. Loading files, reading them, and The operations felt straightforward, and the ecosystem of supporting libraries makes exploratory analysis even easier. In contrast, Spark, which is geared toward distributed processing and designed to handle large volumes of data, required more preparation: configuring the virtual machine, moving files via HDFS, and managing network issues added an extra layer of complexity. Furthermore, some functions that can be accomplished with just a few lines of code in Pandas required more extensive or customized procedures in Spark. This led me to perceive Pandas as a more convenient and agile option for exploratory analysis tasks and basic transformations, especially for datasets that do not exceed local memory capacity.

In my initial experience, working with Spark required more effort because it was a new, distributed environment, where I felt I had to plan technical aspects unrelated to the analysis itself more carefully. Even so, I recognize that its potential to process massive volumes of data in a scalable way is key in larger-scale projects. Ultimately, I believe that both tools serve distinct roles and complement each other: Pandas offers speed and simplicity for working in local environments, while Spark is more robust and powerful for large-scale data processing. Having experimented with both gives me a broader perspective to decide which one is best suited based on the objectives and characteristics of each project.

5. Dashboard on Tableau Public

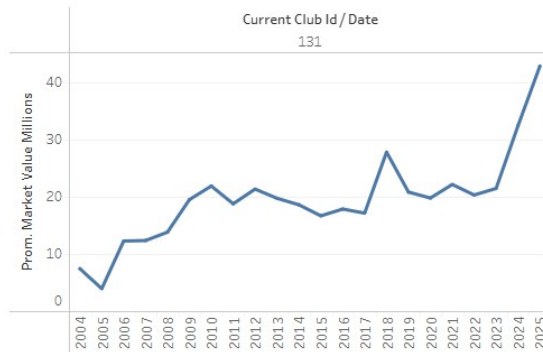
To complement the analysis, an interactive dashboard has been developed to visualize and answer some of the project's key questions. This dashboard was implemented in Tableau Public to facilitate data interpretation and communicate findings in a clear and accessible way to diverse audiences.

Obligatorio Big Data - Dashboard Transfermarket

TOP 10 de equipos con mayor cantidad de partidos



Promedio de valor de mercado de los jugadores del FC Barcelona y su evolución



Mapa mundial con la cantidad de jugadores según su nacionalidad



Link:

https://public.tableau.com/shared/57RCQSWWF?:display_count=nC:origin=viz_share_link

6. Selected Model

To structure the soccer data, we propose implementing a normalized model. This choice is based on the presence of well-defined entities (players, clubs, matches, transfers, etc.) and clear relationships between them, which makes it easier to maintain data integrity, minimize redundancies, and simplify updates. Additionally, normalization offers flexibility to adapt the model to new entities or future metrics without creating duplicates.

This model will help answer questions such as:

- What is a player's average performance across all competitions?
- Which clubs have made the most transfers in a given season?
- How have a player's market valuations evolved over the years?

7. Model structure in Hive

The following structure is proposed in Hive, with external tables stored in HDFS (for example, in CSV format). This approach allows the logical schema to be defined without losing the link to the original data, thereby avoiding file duplication and facilitating integration with other ETL processes. Furthermore, the separation between the logical structure and the physical storage helps to maintain consistency in distributed environments and offers high scalability.

Dimension tables (entities):

- **Players:** personal and athletic attributes (ID, name, nationality, age, position, current club). This will allow for an analysis of the players' career paths and characteristics of the players.
- **Clubs:** identifier, name, country, stadium. Useful for comparing club characteristics or filtering statistics by country or competition.

- **Competitions:** information on leagues or tournaments (ID, name, type, confederation). This supports performance analysis by tournament.
- **Matches:** match data (ID, competition, date, teams involved, result). This will be key for tracking results and analyzing performance over time.

Fact tables (transactional):

- **Appearances:** links players to matches, detailing goals, assists, cards, minutes played. Facilitates analysis of individual and team performance.
- **Match events:** records key actions (goals, substitutions, cards). This allows for a deeper analysis of decisive moments in the game.
- **Lineups:** indicates which players participated and in which position. This will be useful for evaluating coaches' tactical decisions.
- **Transfers:** moves between clubs, with date and amount. This allows you to study the transfer market and identify buying or sellers.
- **Player valuations:** tracks market value over time, useful for tracking development and comparing players.

These tables will be linked via primary and foreign keys:

- Players are associated with current and former clubs
- Matches are linked to clubs and competitions
- Appearances and events are linked to matches and players
- Transfers link players to their source and destination clubs

8. Conclusions

This analysis allowed for a deeper exploration of soccer data obtained from Transfermarkt, answering a series of key questions about the performance of players, clubs, and competitions. Through a process of data cleaning and transformation using primarily Pandas in the first stage, and then replicating the analysis with Spark, we were able to structure the information efficiently and comprehensibly, laying the groundwork for drawing relevant conclusions about the dynamics of professional soccer in recent years.

In terms of tools, the experience with Pandas proved particularly positive for conducting exploratory analyses and basic transformations, thanks to its ease of use. In contrast, Spark required more preparation and additional knowledge related to the configuration of distributed environments, but it also demonstrated their potential for processing large volumes of data in a scalable and efficient manner, which broadens the scope for future projects of greater magnitude. This exercise provided a more comprehensive understanding of when to apply each technology based on the objectives and the size of the dataset.

Likewise, the creation of interactive visualizations in Tableau Public enriched the interpretation of the results, demonstrating the value of dashboards as a tool for communicating findings to non-technical users.

From a conceptual standpoint, the proposal for a data model Standardized in Hive, with storage in HDFS and external tables, this reinforces the importance of maintaining integrity, scalability, and flexibility in distributed data management—fundamental aspects in Big Data environments.

Regarding the questions raised, although some focused on relatively simple metrics, the analysis provided significant insights into performance patterns, transfer flows, market value trends, market values, and trends in roster rejuvenation across different leagues and

teams. This lays a solid foundation for addressing more complex questions in future stages and making greater use of Big Data capabilities.

Throughout this project, I sought to strengthen and expand the knowledge , incorporating best practices and approaches focused on professional development, with the goal of preparing myself as best as possible for future career challenges. This commitment to learning and continuous improvement was key to tackling the challenges of this analysis with greater confidence and perspective.

In summary, carrying out this work allowed me not only to consolidate valuable technical skills in the use of data analysis and processing tools, but also to develop a critical perspective on the business questions that can be answered and to understand the practical challenges of working with in distributed environments. This learning represents an important step toward tackling larger-scale data analysis projects with greater confidence and preparedness.